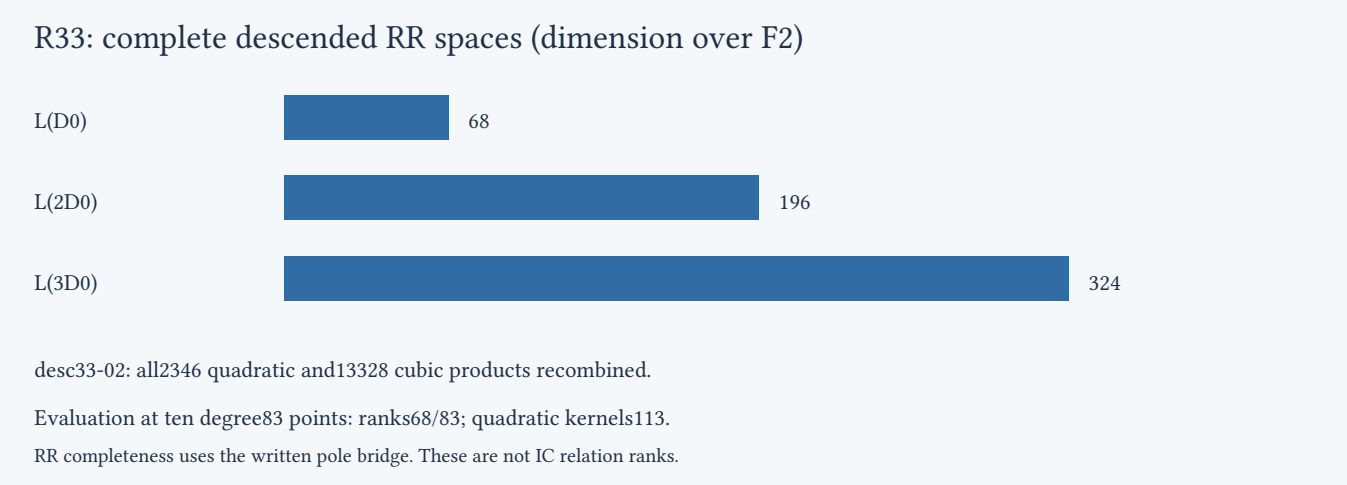
Koblitz cover transfers — R33

Observations and certificate boundaries; no scientific promotion

The fixed complete source basis is descended through the actual characteristic2 semilinear involution. All68 rational functions are explicit over F2. The native operator checks include $S^2=I$, $N^2=0$, $C^2=I$ and the descent cocycle; corrupted controls are rejected. Multiplication coordinates and every basis recombination are retained. Completeness uses the inherited R27 proper pole and Riemann–Roch bridge, not a new coefficient-box search.



All ten degree83 twist points pass physical Artin–Schreier identification, original-function/deck/descent compatibility and exact Gaussian checks. The native computation completed once; a wrapper's shared-directory tar failed while another run was writing. Its separate reconciliation preserves the native exit, marker, error scan and all ten checks. The math was not repeated.

The exhaustive ordinary F32 search finds two distinct closed degree5 points. Their actual T^2 action is coefficient Frobenius, sigma exchanges the points, and the integer projector cancels their closed divisors. This supplies a proposed rational padding9Q, not a checked nonzero desired-factor class.

The2150 independent F2 twist quadrics now serialize and round-trip to the certified kernel. The full four-variable padding Jacobian has rank3 and determinant0 at both points, still without a local-dimension proof.

Local geometry and failures

The small4x3 Jacobian has unit maximal minors at both padding points; the generic source algebra has dimension32 over F2(u), with exact two-way lifts to its six-generator Groebner basis. The old51-product syzygy presentation failed at a padding branch. Rank is therefore kept separate from a valid local presentation. The expanded degree83 checker reached its original 1200second cap without a completed point result. The direct quotient run also capped at1200seconds: three completed omissions retain10/10/7 generators with exact lifts, while omission4 is unfinished. This is partial evidence, not a successful full producer. Denominator-checked specialization remains separately gated.

A failed native run with exit0 and a trailing marker is rejected whenever Singular reports errors. Failed identity initialization, reserved names, ideal construction and F4/F2 input framing are retained. No failure or cap is interpreted as a mathematical impossibility result. The run index below states the terminal or currently pending state of each attempt.

Formal scope and next construction

Six named universal Lean theorems (seven audited declarations) establish characteristic2 descent algebra and unit-coefficient equation elimination. Both modules compile and pass transitive axiom audits with zero failing or trusted declarations. Actual native matrices, fields, geometry, genus and RR completeness are not imported into Lean.

Next gates are a source-bound local presentation, proper normalized branch binding, complete nine-jets at Q, then all ten coupled divisors P+9Q. The standard Khuri-Makdisi large model requires L(3D0), L(6D0) and a checked multiplication map. The proposed missing L(6D0) dimension is708; no completed Picard operation or intended order11 factor evaluation is claimed.

Sources read: Stacks35.6 and29.35.14; Singular official reference.doc quotient/lift; Khuri-Makdisi math/0409209v3 sections2 and3. No IC timing, independent scientific replay, unknown-scalar recovery, full saturation or proper polarized factor embedding. Native4GiB/1200s each; pinned Lean4.34.1/mathlibd13f23b, one job/4096MiB/1200s. Full root suite not passed.

Attempt inventory

- desc33-01: rejected
- desc33-02: accepted
- eval33-01: accepted
- lean-local33-01: accepted
- lean33-01: rejected
- lean33-02: accepted
- local33-check83-01: capped
- local33-compact-check5-01: rejected
- local33-compact-dump01: accepted
- local33-diagnostic01: rejected
- local33-diagnostic02: accepted
- local33-function01: accepted
- local33-generate01: accepted
- local33-quotient01: capped
- pad33-01: accepted
- padding-deck33-01: rejected
- padding-deck33-02: accepted
- padding33-tangent01: accepted
- quadrics33-01: rejected
- quadrics33-02: accepted